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United States Patent Application Publication

20250281938

Kind Code

A1

Publication Date

September 11, 2025

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SPIT REDUCTION FOR FLUID APPLICATORS

Abstract

A fluid applicator includes a gun having a trigger, a valve assembly moveable in a first direction by actuation of the trigger, a first spring that biases the valve assembly in a second direction and is compressible by movement of the valve assembly in the first direction, and a second spring that stores energy in a first phase of the movement of the valve assembly in the first direction and releases the stored energy in a second phase of the movement of the valve assembly in the first direction. The applicator includes a spray tip assembly having a spray tip, a tip saddle, and a seal element disposed within the tip saddle and having a top surface in contact with an exterior surface of the tip saddle, the seal element defining an inlet to the spray tip assembly and forming a seal between the gun and the spray tip assembly.

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Family ID: 96948363

Appl. No.: 19/032961

Filed: January 21, 2025

Related U.S. Application Data

us-provisional-application US 63562986 20240308

Publication Classification

Int. Cl.: B05B1/30 (20060101); B05B9/01 (20060101); B05B9/04 (20060101)

U.S. Cl.:

CPC B05B1/3046 (20130101); B05B9/01 (20130101); B05B9/0403 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application is based on and claims the benefit of U.S. provisional patent application Ser. No. 63/562,986, filed Mar. 8, 2024, the content of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] In one example of a fluid application system, a pump receives and pressurizes a fluid, delivers the pressurized fluid to a fluid applicator, which, in turn, applies the pressurized fluid to a surface using a spray tip having a geometry selected to emit a desired spray pattern (e.g., a round pattern, a flat pattern, or a fan pattern, etc.). The fluid may comprise any fluid applied to surfaces, including, but not limited to, for example, paint, primer, lacquers, foams, textured materials, plural components, adhesive components, etc.

[0003] One example of a fluid applicator is a spray gun. Spray guns include a gun and a spray tip assembly. The spray gun generally includes an inlet, that receives fluid from a fluid source (e.g., fluid pumped by a pump), and an outlet. A spray tip assembly is coupled to the gun at the outlet of the spray gun. The spray tip assembly includes a spray tip, along with other items. The spray tip includes an inlet, that receives fluid from the gun, and an outlet that releases the fluid in a desired spray pattern, such as a spray fan. The spray tip breaks up, or atomizes, the fluid for delivery in the desired spray pattern. A spray gun can also include a valve that is actuatable, such as by actuation of a trigger of the spray gun, to selectively allow fluid to flow to the spray tip.

[0004] While examples described herein are in the context of applying paint to a surface, it is understood that the concepts are not limited to these particular applications. As used herein, paint includes substances composed of coloring matter, or pigments, suspended in a liquid medium as well as substances that are free of coloring matter or pigment. Paint may also include preparatory coatings, such as primers, and can be opaque, transparent, or semi-transparent. Some particular examples include, but are not limited to, latex paint, oil-based paint, stain, lacquers, varnishes, inks, etc.

[0005] The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter.

SUMMARY

[0006] One example fluid application system includes one or more pumps, a fluid delivery line coupled to the one or more pumps, and a fluid applicator. The fluid applicator includes a gun; and a spray tip assembly removably coupled to the gun. The spray tip assembly includes a spray tip, a tip saddle, and a seal disposed, at least partially, within the tip saddle and having a top surface in contact with an exterior surface of the tip saddle, the seal defining an inlet to the spray tip assembly and forming a seal between the gun and the spray tip assembly. In one example, the spray tip assembly further includes a retaining element, the tip saddle and the spray tip disposed, at least partially, within the retaining element and the seal comprises a captivating wing configured to contact an inner surface of the retaining element and thereby establish a friction fit between the seal and the retaining element. In one example, the seal comprises a hard polymer, such as, but not limited to, nylon.

[0007] One example fluid application system includes one or more pumps, a fluid delivery line coupled to the one or more pumps, and a fluid applicator. The fluid applicator includes a spray tip assembly and a gun, removably coupled to the spray tip assembly. The gun includes a trigger; and a valve assembly configured to be moved in a first direction by actuation of the trigger. The valve assembly includes a first spring configured to bias the valve assembly in a second direction and to be compressed by movement of the valve assembly in the first direction; and a second spring configured to store energy in a first phase of the movement of the valve assembly in the first

direction of movement and to release the stored energy in a second phase of the movement of the valve assembly in the first direction to assist in moving the valve assembly in the first direction. In one example, the second spring is a compression spring. In one example, the second spring is a tension spring. In one example, both the first spring and the second spring are configured to contact fluid applied by the fluid application system. In one example, the gun further includes a spring chamber, a valve fluid volume, and a seal configured to fluidically seal the spring chamber from the valve fluid volume. In one example, both the first spring and the second spring are disposed in the spring chamber. In one example, the first spring is disposed in the spring chamber and the second spring is disposed in the valve fluid volume. In one example, the second spring is in contact with the trigger.

[0008] One example fluid application system includes one or more pumps, a fluid delivery line coupled to the one or more pumps, and a fluid applicator. The fluid applicator includes a gun. The gun includes a trigger; and a valve assembly configured to be moved in a first direction by actuation of the trigger. The valve assembly includes a first spring configured to bias the valve assembly in a second direction and to be compressed by movement of the valve assembly in the first direction; and a second spring configured to store energy in a first phase of the movement of the valve assembly in the first direction of movement and to release the stored energy in a second phase of the movement of the valve assembly in the first direction to assist in moving the valve assembly in the first direction. The fluid applicator includes a spray tip assembly removably coupled to the gun. The spray tip assembly includes a spray tip, a tip saddle, and a seal disposed, at least partially, within the tip saddle and having a top surface in contact with an exterior surface of the tip saddle, the seal defining an inlet to the spray tip assembly and forming a seal between the gun and the spray tip assembly. In one example, the spray tip assembly further includes a retaining element, the tip saddle and the spray tip disposed, at least partially, within the retaining element and the seal comprises a captivating wing configured to contact an inner surface of the retaining element and thereby establish a friction fit between the seal and the retaining element. In one example, the seal comprises a hard polymer, such as, but not limited to, nylon. In one example, the second spring is a compression spring. In one example, the second spring is a tension spring. In one example, both the first spring and the second spring are configured to contact fluid applied by the fluid application system. In one example, the gun further includes a spring chamber, a valve fluid volume, and a seal configured to fluidically seal the spring chamber from the valve fluid volume. In one example, both the first spring and the second spring are disposed in the spring chamber. In one example, the first spring is disposed in the spring chamber and the second spring is disposed in the valve fluid volume. In one example, the second spring is in contact with the trigger.

[0009] One example spray tip assembly, configured to be removably coupled to a gun and to emit fluid in a spray pattern, includes a spray tip, a tip saddle, and a seal disposed, at least partially, within the tip saddle and having a top surface in contact with an exterior surface of the tip saddle, the seal defining an inlet to the spray tip assembly and forming a seal between the gun and the spray tip assembly. In one example, the spray tip assembly further includes a retaining element, the tip saddle and the spray tip disposed, at least partially, within the retaining element and the seal comprises a captivating wing configured to contact an inner surface of the retaining element and thereby establish a friction fit between the seal and the retaining element. In one example, the seal comprises a hard polymer, such as, but not limited to, nylon.

[0010] One example fluid gun, configured to be removably coupled to a spray tip and to control a flow of fluid to the spray tip, includes a trigger and a valve assembly. The gun includes a trigger; and a valve assembly configured to be moved in a first direction by actuation of the trigger. The valve assembly includes a first spring configured to bias the valve assembly in a second direction and to be compressed by movement of the valve assembly in the first direction; and a second spring configured to store energy in a first phase of the movement of the valve assembly in the first direction of movement and to release the stored energy in a second phase of the movement of the

valve assembly in the first direction to assist in moving the valve assembly in the first direction. In one example, the second spring is a compression spring. In one example, the second spring is a tension spring. In one example, both the first spring and the second spring are configured to contact the fluid. In one example, the gun further includes a spring chamber, a valve fluid volume, and a seal configured to fluidically seal the spring chamber from the valve fluid volume. In one example, both the first spring and the second spring are disposed in the spring chamber. In one example, the first spring is disposed in the spring chamber and the second spring is disposed in the valve fluid volume. In one example, the second spring is in contact with the trigger.

[0011] One example fluid applicator includes a gun and a spray tip assembly removably coupled to the gun. The spray tip assembly includes a spray tip, a tip saddle, and a seal disposed, at least partially, within the tip saddle and having a top surface in contact with an exterior surface of the tip saddle, the seal defining an inlet to the spray tip assembly and forming a seal between the gun and the spray tip assembly. In one example, the spray tip assembly further includes a retaining element, the tip saddle and the spray tip disposed, at least partially, within the retaining element and the seal comprises a captivating wing configured to contact an inner surface of the retaining element and thereby establish a friction fit between the seal and the retaining element. In one example, the seal comprises a hard polymer, such as, but not limited to, nylon.

[0012] One example fluid applicator includes a spray tip assembly and a gun, removably coupled to the spray tip assembly. The gun includes a trigger; and a valve assembly configured to be moved in a first direction by actuation of the trigger. The valve assembly includes a first spring configured to bias the valve assembly in a second direction and to be compressed by movement of the valve assembly in the first direction; and a second spring configured to store energy in a first phase of the movement of the valve assembly in the first direction of movement and to release the stored energy in a second phase of the movement of the valve assembly in the first direction to assist in moving the valve assembly in the first direction. In one example, the second spring is a compression spring. In one example, the second spring is a tension spring. In one example, both the first spring and the second spring are configured to contact fluid applied by the fluid applicator. In one example, the gun further includes a spring chamber, a valve fluid volume, and a seal configured to fluidically seal the spring chamber from the valve fluid volume. In one example, both the first spring and the second spring are disposed in the spring chamber. In one example, the first spring is disposed in the spring chamber and the second spring is disposed in the valve fluid volume. In one example, the second spring is in contact with the trigger.

[0013] One example fluid applicator includes a gun. The gun includes a trigger; and a valve assembly configured to be moved in a first direction by actuation of the trigger. The valve assembly includes a first spring configured to bias the valve assembly in a second direction and to be compressed by movement of the valve assembly in the first direction; and a second spring configured to store energy in a first phase of the movement of the valve assembly in the first direction of movement and to release the stored energy in a second phase of the movement of the valve assembly in the first direction to assist in moving the valve assembly in the first direction. The fluid applicator includes a spray tip assembly removably coupled to the gun. The spray tip assembly includes a spray tip, a tip saddle, and a seal disposed, at least partially, within the tip saddle and having a top surface in contact with an exterior surface of the tip saddle, the seal defining an inlet to the spray tip assembly and forming a seal between the gun and the spray tip assembly. In one example, the spray tip assembly further includes a retaining element, the tip saddle and the spray tip disposed, at least partially, within the retaining element and the seal comprises a captivating wing configured to contact an inner surface of the retaining element and thereby establish a friction fit between the seal and the retaining element. In one example, the seal comprises a hard polymer, such as, but not limited to, nylon. In one example, the second spring is a compression spring. In one example, the second spring is a tension spring. In one example, both the first spring and the second spring are configured to contact fluid applied by the fluid applicator. In

one example, the gun further includes a spring chamber, a valve fluid volume, and a seal configured to fluidically seal the spring chamber from the valve fluid volume. In one example, both the first spring and the second spring are disposed in the spring chamber. In one example, the first spring is disposed in the spring chamber and the second spring is disposed in the valve fluid volume. In one example, the second spring is in contact with the trigger.

[0014] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the background.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 is a perspective view showing one example fluid application system.

[0016] FIG. 2 is a side view showing one example fluid applicator.

[0017] FIG. 3 is a perspective view showing one example spray tip.

[0018] FIG. 4 is a partial front view showing one example spray tip.

[0019] FIGS. 5A-5C (collectively referred to herein as FIG. 5) are sectional views showing one example fluid applicator.

[0020] FIGS. 6A-6C (collectively referred to herein as FIG. 6) are sectional views showing one example fluid applicator.

[0021] FIG. 7 is a schematic illustration of one example valve assembly.

[0022] FIG. 8 is a graphical illustration.

[0023] FIG. 9 is a sectional view showing one example fluid applicator.

[0024] FIG. 10 is simplified pictorial illustration showing one example valve assembly.

[0025] FIGS. 11A-11B (collectively referred to herein as FIG. 11) are sectional views showing one example fluid applicator.

[0026] FIGS. 12A-12B (collectively referred to herein as FIG. 12) are sectional views showing one example fluid applicator.

[0027] FIG. 13A is sectional to view showing one example valve assembly.

[0028] FIG. 13B is perspective view showing one example valve assembly.

[0029] FIG. 14 is a schematic illustration of one example valve assembly.

[0030] FIG. 15 is a schematic illustration of one example fluid applicator.

[0031] FIG. 16 is a schematic illustration of one example fluid applicator.

[0032] FIG. 17 is a schematic illustration of one example fluid applicator.

[0033] FIG. 18 is a schematic illustration of one example fluid applicator.

[0034] FIG. 19 is a schematic illustration of one example fluid applicator.

[0035] FIG. 20 is a block diagram showing one example fluid application system.

DETAILED DESCRIPTION OF THE DRAWINGS

[0036] For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the examples illustrated in the drawings, and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the disclosure is intended. Any alterations and further modifications to the described devices, systems, methods, and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one example may be combined with the features, components, and/or steps described with respect to other examples of the present disclosure.

[0037] As spray guns are activated and deactivated (e.g., as a valve is opened and closed), spitting can occur. Spitting is the expulsion of droplets or uneven bursts of fluid instead of in a smooth and even pattern. Spitting can result in inconsistent finish on the target surface, application of fluid on non-target surfaces, wasted fluid, and additional labor for touch-up. There are two forms of spitting, start spitting (or starting spit) and stop spitting (or stopping spit). Start spitting occurs when the gun is activated (e.g., valve is opened) and results from too slow a pressurization of the fluid volume downstream of the valve (which includes the fluid volume of the spray tip as well as other items). Stop spitting occurs when the gun is deactivated (e.g., valve is closed) and results from too slow a depressurization of the fluid volume downstream of the valve.

[0038] A number of factors influence the occurrence and severity of spitting, such as the size of the fluid volume downstream of the valve, how quickly the valve can open and close, the pressure accumulation of the fluid volume downstream of the valve, as well as other factors.

[0039] The present description proceeds with example systems and methods for reducing spitting in fluid applicators. The example systems and methods provide for one or more of reducing the size of the fluid volume downstream of the valve (without necessarily reducing the fluid volume of the spray tip), increasing the speed at which the valve can actuate (e.g., open or close, or both), and decreasing the pressure accumulation of the fluid volume downstream of the valve. Each of reducing the size of the fluid volume downstream of the valve, increasing the speed at which the valve can actuate (e.g., open or close, or both), and decreasing the pressure accumulation of the fluid volume downstream of the valve can reduce spitting.

[0040] FIG. 1 is a perspective view showing one example fluid application system **1**. Fluid application system **1**, illustratively shown as an airless fluid spraying system (e.g., a high efficiency airless spraying system), includes pump **2** that is mounted on a cart **4** and couples to applicator **10** through fluid delivery line **6** (e.g., a hose). Pump **2** includes a fluid intake **8** that is disposed within a fluid source (e.g., a five-gallon bucket of paint). Pump **2** pumps the fluid from the fluid source through fluid intake **8** and pumps the fluid at a given pressure to applicator **10** through fluid delivery line **6** (e.g., a flexible hose). In one example, pump **2** can pressurize the fluid between 1500-3500 PSI.

[0041] FIG. 2 is a side view showing an example applicator **10**. Applicator **10** is used in a fluid spraying system (e.g., fluid application system **1**) to apply fluid to a surface (e.g., apply paint to a wall). The fluid enters through inlet **20**, and exits from outlet **50**, after passing through a fluid channel (not explicitly shown) within applicator **10**. Fluid inlet **20** may be coupled to a fluid delivery line, such as fluid delivery line **6**. Tip **32** is coupled to applicator **10** and has an outlet **50**. Tip **32** often is reversible (e.g., tip **32** can be rotated around its longitudinal axis such that the inlet and outlet are flipped in position (i.e., inlet of tip **32** facing away from applicator **10** and outlet of tip **32** facing towards applicator **10**)) or removable from applicator **10**. The reversibility of spray tip **32** can help with cleaning.

[0042] FIG. 3 is a perspective view showing an example spray tip **32**. Spray tip **32** includes flag **33**, tip stem **34**, and receiving channel **36**. Flag **33** can be coupled to tip stem **34** in various ways including, for example, but not by limitation, press fitting flag onto tip stem **34** or over molding flag **33** onto tip stem **34**. Flag **33** provides a convenient surface for handling spray tip **32**, particularly when spray tip **32** is installed in an applicator and can be used to indicate the directionality of spray tip **32**. Flag **33** can comprise various materials, for example, polymer. Tip stem **34** can comprise various materials, for example, metal such as stainless steel. A receiving channel **36** can be provided through tip stem **34**, such as by machining, cutting, etc. The receiving channel **36** extends a distance between a front **38** of spray tip **32** and a rear (or back) **40** of spray tip **32**. In some examples, the receiving channel **36** may extend from a front **38** of spray tip **32** to a rear **40** of spray tip **32** and yet, in other examples, the receiving channel **36** may extend some other distance. The receiving channel **36** will be shown in more detail below.

[0043] FIG. 4 is a partial front view showing example spray tip **32**. As illustrated in FIG. 4, a tip

piece **60** can be placed and retained within receiving channel **36**. Additionally, as will be shown below, a pre-orifice element can, optionally, be placed in the receiving channel **36**. Examples of spray tip **32** are shown below (e.g., spray tip **132** and spray tip **432**).

[0044] FIGS. 5A-C (collectively referred to herein as FIG. 5) are sectional views showing one example fluid applicator **100**, illustratively a spray gun. Fluid applicator **100** can be used in a fluid application system, such as fluid application system **1** shown in FIG. 1, in place of fluid applicator **10**.

[0045] Spray gun **100** includes gun (or gun body) **110**, inlet **120**, trigger **126**, spray tip assembly **130**, outlet **150**. Gun **110** includes, among other things, valve fluid volume **112**, valve spring chamber **114**, inlet fluid volume **122**, handle **128**, valve assembly **160**, outlet **181**, and can include various other items.

[0046] Inlet **120** is coupled to a delivery line (e.g., **6**) and is configured to receive fluid pumped through the delivery line (e.g. **6**). Fluid enters gun **110** through inlet **120** and travels through inlet fluid volume **122**, disposed within handle **128**, to valve fluid volume **112**.

[0047] Spray tip assembly **130** includes spray tip **132**, outlet guard **134**, tip saddle **136**, seal **138**, retaining element **140**, and coupling mechanism **142**. Spray tip **132** can be similar to and used in place of spray tip **32**. Spray tip **132** includes tip piece **133** and pre-orifice piece **135**. In other examples, spray tip **132** could include, as a pre-orifice element, a pre-orifice formed in the body of spray tip **132** rather than a pre-orifice element as a separate pre-orifice piece, such as pre-orifice piece **135**. In other examples of spray tip **132**, a pre-orifice element, such as a pre-orifice piece **135**, need not be included. Tip piece **133** defines outlet **150** and is configured to emit fluid in a desired spray pattern. Spray tip **132**, tip saddle **136**, and seal **138** are disposed within retaining element **140** and are thereby coupled to one another. Spray tip **132** is rotatable (about its longitudinal axis) within retaining element **140** between a normal operating position (shown in FIG. 5, where outlet **150** is facing away from gun **110**) and a cleaning operating position (where outlet **150** is facing toward gun **110**). Outlet guard **134** is coupled to retaining element **140**, such as by a snap-on connection. Coupling mechanism **142** is coupled to retaining element **140**, such as by a snap-on connection. Coupling mechanism **142** is also coupled to gun body **110** (threaded connection in the illustrated example) and thereby couples spray tip assembly **130** to gun **110**.

[0048] Seal **138** creates a seal between gun **110** and spray tip assembly **130**. As shown in FIG. 5 (best seen in FIG. 5B), seal **138** includes a top surface **151** in contact with an interior surface **152** of retaining element **140**, a downstream surface **153** and a bottom (or interior) surface **154** in contact with an exterior surface **155** of tip saddle **136**, and an upstream surface **156** in contact with an exterior surface **157** of gun **110**.

[0049] Valve assembly (or valve) **160** includes valve seat **162**, valve ball **164**, valve shaft **166**, spring plate **168**, spring **170**, and seal **172**, and can include various other items. Valve assembly **160** is moveable, in the directions **190** and **192**, by actuation of the trigger **126** (as described below), between a seated (or closed) position (as shown in FIG. 5) and a plurality of open positions (from initially open to fully open). It will be appreciated by those skilled in the art that there is a range of open positions of valve assembly **160**, the range extending between an initial point when valve ball **164** is retracted sufficiently to initially allow some fluid through valve seat **162** to spray tip assembly **130**, though the valve ball **164** is still disrupting flow through the valve seat **162**, and a fully open (or fully retracted) point in which valve assembly **160** is retracted to the fullest extent and the valve ball **164** is fully pulled away from valve seat **162**. In other words, it will be understood that the valve assembly **160** can effectively control the flow rate of fluid to the fluid volume of spray tip assembly **130**. Valve ball **164**, valve shaft **166**, and spring plate **168** are coupled together.

[0050] Trigger **126** is moveably coupled to the gun body **110** and is moveable, such as by application and release (or reduction) of force from user's hand, in directions **190** and **192**, between a deactivated (or resting) position (shown in FIG. 5) and an activated position. Trigger **126** is

coupled to spring plate **168** such that actuation of trigger **126** in the direction **190** causes spring plate **168** to move in the direction **190** and thereby compress spring **170** to open valve assembly **160**. When valve assembly **160** is opened, fluid flows from valve fluid volume **112**, through seat **162**, out of gun outlet **181** to spray tip assembly **130** through spray tip assembly inlet **183**.

[0051] Spring **170** biases valve assembly **160** in a closed position, and is compressible, as described above, to open valve assembly **160**. Hydrostatic pressure of fluid in valve chamber **112** also biases valve **160** in a closed position. The biasing force of the spring **170** and of the hydrostatic pressure can be overcome by application of force to trigger **126**.

[0052] Seal **172** operably seals spring chamber **114** from valve fluid volume **112**. In this way, spring plate **168** and spring **170** are not in contact with fluid, which can improve wear (or longevity) of spring plate **168** and spring **170**. Further, the seal between valve fluid volume **112** and spring chamber **114** can even allow for access to spring plate **168** and spring **170**, such as for maintenance or replacement, even when the gun **110** is pressurized. As shown, applicator **100** includes a spring chamber cap **173** that is removably coupled (via threaded connection as shown) to gun body **110**. Spring chamber cap **173** can be removed to allow access to spring chamber **114**. Spring chamber cap **173** also provides a surface against which spring **170** bears to be compressed and against which spring **170** bears to extend.

[0053] FIGS. **6A-C** (collectively referred to herein as FIG. **6**) are sectional views showing one example fluid applicator **200**, illustratively a spray gun. Fluid applicator **200** can be used in a fluid application system, such as fluid application system **1** shown in FIG. **1**, in place of fluid applicator **10**. Fluid applicator **200** is similar to fluid applicator **100** shown in FIG. **5** and thus, similar items are numbered similarly.

[0054] It can be seen in FIG. **6** that fluid applicator **200** includes a spray tip assembly **230** and a valve assembly **260**. Spray tip assembly **230** differs from spray tip assembly **130** in that spray tip assembly **230** includes tip saddle **236** and seal **238** (illustratively **238-1**), different from tip saddle **136** and seal **138** of spray tip assembly **130**. Valve assembly (or valve) **260** differs from valve assembly (or valve) **160** in that valve assembly **260** includes a spring plate **268**, different from spring plate **168**, and also includes a spring clip **274** and an additional (or second) spring **276**.

[0055] Tip saddle **236** and seal **238** provide for a reduction of the volume downstream of the valve, without changing the spray tip **132** or the tip piece **133** or pre-orifice piece **135**, and for a reduction of the pressure accumulation of the volume downstream of the valve. Seal **238** also defines an inlet **283** of spray tip assembly **230**. As can be seen in FIG. **6**, tip saddle **236** includes a seal-recess (or seal seat) **237** in which seal **238** is disposed. Seal **238**, like seal **138**, creates a seal between gun **110** and the spray tip assembly. However, seal **238** is of a smaller size (smaller diameter, both interior and exterior diameter) than seal **138**. Additionally, the elasticity of the volume downstream of the valve is reduced, and thus the pressure accumulation of the volume downstream of the valve is reduced, by the arrangement of seal **238**. As shown, seal **238** includes a top surface **251** in contact with an exterior surface **255** of tip saddle **236**. The contact between top surface **251** and exterior surface **255** allows seal **238** to still provide sealing between gun **110** and the spray tip assembly **260**, at a smaller size than that of seal **138**, while also providing resistance the expansion of seal **238** to reduce or prevent pressure accumulation. Seal **238** also includes a downstream surface **253** in contact with the exterior surface **255** of tip saddle **236**. As can be seen, seal **238** also includes an upstream surface **256** in contact with the exterior surface **157** of gun **110**. Further, as can be seen, seal **238** includes a bottom (or interior) surface **254** that does not contact another item of spray tip assembly **260** (whereas the bottom surface **154** of seal **138** contacts the exterior surface **155** of tip saddle **136**). Another example of seal **238** will be shown in FIG. **8**.

[0056] The reduced size and reduced pressure accumulation of the volume downstream of the valve assembly **260**, will help to reduce spitting, both start spitting and stop spitting. The reduced size and pressure accumulation of the volume downstream of the valve assembly **260** will pressurize more quickly (e.g., more quickly than the volume downstream of the valve assembly **160**) and thus

reduce start spitting. The reduced size and pressure accumulation of the volume downstream of the valve assembly **260** will depressurize more quickly (e.g., more quickly than the volume downstream of the valve assembly **160**) and thus reduce stop spitting.

[0057] As shown in FIG. **6**, valve assembly **260** includes an additional spring **276** disposed around valve shaft **166** and disposed between spring plate **268** and a spring clip **274**. Movement of spring plate **268**, in the direction **190**, by way of actuation of trigger **126**, will, eventually, move spring clip **274**, and thus valve shaft **166**, in the direction **190**.

[0058] As can be seen, there is an adjustable gap **278** between spring plate **268** and spring clip **274** that closes with actuation of spring plate **268** in the direction **190**. In one example, the gap **278**, in a valve closed position (as shown in FIG. **6**), is 0.03 inches or 0.75 millimeters. Though, in other examples, the gap **278**, in a valve closed position, could be larger or smaller. As spring plate **268** is moved in the direction **190**, shoulder **267** of spring plate **268** is moved along wall **277** of spring clip **274**, in the direction **190**, compressing spring **276** against shoulder **279** of spring clip **274** and storing energy in spring **276**. Additionally, spring plate **268**, during this initial phase of movement (closing the gap **278** phase), partially compresses spring **170**. During this initial phase of movement, the closing of the gap phase, the spring clip **274**, and thus the valve shaft **166**, remains stationary or, at least, does not move in the direction **190**, and the ball **164** is not dislodged from valve seat **162**. Thus, it will be understood that that the double spring system shown in FIG. **6** allows for an amount (e.g., 0.03 inches or 0.75 millimeters or other gap sizes) of compression before the valve ball **164** is dislodged from the seat **162**. That is, the double spring system shown in FIG. **6** allows for an amount (e.g., 0.03 inches or 0.75 millimeters or other gap sizes) of movement of spring plate **268** before movement of spring clip **274** and thus valve shaft **166** and valve ball **164**. Eventually, with continual movement in the direction **190**, the gap **278** is closed. In a subsequent phase of movement, when the gap is closed, such that shoulder **269** of spring plate **268** contacts shoulder **275** of spring clip **274**, spring clip **274** will begin moving in the direction **190** and will be assisted in the movement in direction **190** by the stored energy of spring **276** which will assist in opening the valve (unseating the valve ball **164**) more quickly. This subsequent phase of movement, wherein the spring **276** releases its stored energy, includes an initial opening of the valve assembly **260** to a fully open position of the valve assembly **260**. That is, the valve ball **164** will become dislodged from the valve seat **162** during this subsequent phase.

[0059] The initial phase of movement (e.g., closing the gap phase) can also be referred to as the energy storage phase. The subsequent phase of movement can also be referred to as the energy release phase.

[0060] The action of the multiple springs shown in FIG. **6**, described above, allows the valve assembly **260** to open more quickly, specifically, to open more quickly from initially open to fully open and thereby to pressurize the volume downstream of the valve **260** more quickly. The valve assembly **260** opening more quickly (e.g., more quickly than the valve assembly **160**) will help to reduce spitting (e.g., start spitting) by allowing for the volume downstream of the valve to be pressurized more quickly.

[0061] Additionally, while the example shown in FIG. **6** shows that a spray gun **200** includes both spray tip assembly **230** and valve assembly **260**, it will be understood that in other examples, a spray gun could include one or the other. For example, a spray gun could include spray tip assembly **230** but include a different valve assembly, such as valve assembly **160**. In another example, a spray gun could include valve assembly **260** but include a different spray tip assembly, such as spray tip assembly **130**. As discussed, the reduced pressure accumulation downstream of the valve, provided by spray tip assembly **230**, can reduce spitting and the quicker opening of the valve assembly **260** can reduce spitting.

[0062] FIG. **7** is a schematic illustration of valve assembly **260**. As shown in FIG. **7**, actuation of trigger **126** in direction **190** causes spring plate **268** to move in the direction **190**. As illustrated in FIG. **7**, trigger **126** is coupled to spring plate **268** by way of a coupling mechanism **177**. During the

initial phase of movement, spring plate **268** compresses spring **170** and spring **276**, storing energy in spring **276**, until adjustable gap **278** is closed whereby spring plate **268** contacts spring clip **274** and, in a subsequent phase of movement, drives spring clip **274**, and thus, valve shaft **166**, in the direction **190**, opening valve. In the subsequent phase of movement, the stored energy of spring **276** is released and assists in driving spring clip **274**, and thus valve shaft **276**, in the direction **190** to open the valve more quickly.

[0063] FIG. **8** is a graphical illustration showing an improved rate of valve opening provided by the double spring system shown in FIG. **6**. As shown in FIG. **8**, graph **300** include Y axis **302**, X axis **304**, line **306**, line **308**, valve closed **310**, valve initial open **312**, valve fully open **314**, T1, T2, T3, T4, valve opening time **316**, and valve opening time **318**. Y axis **302** comprises a scale of valve open area values and includes, as shown, three illustrated values of valve closed **310**, valve initially open **312**, and valve fully open **314**. Line **306** corresponds to a valve assembly, such as valve assembly **160**, and shows a time it takes the corresponding valve assembly (e.g., **160**) to fully open from an initial opening. Line **380** corresponds to a valve assembly such as valve assembly **260**, and shows a time it takes valve assembly **260** to fully open from an initial opening. X axis **304** comprises a scale of time values and includes, as shown, three illustrated values of T1 (indicating the time when the trigger is initially actuated), T2 (indicating the time when the valve assembly corresponding to line **306** initially opens), T3 (indicating the time when valve assembly **360** corresponding to line **308** initially opens), and T4 (indicating the time when the valve assemblies are fully open).

[0064] As can be seen, the valve assembly associated with line **306** (e.g., valve assembly **160**) initially opens (as indicated by T2) more quickly than valve assembly **260** initially opens (as indicated by T3). This is because the valve assembly **260** allows for an amount of movement of the trigger without opening the valve, as previously discussed. However, as can be seen, the valve assembly associated with line **306** takes longer to fully open from an initial opening, as indicated by valve open time **316** representing an amount of time between T2, when the valve assembly associated with line **306** initially opens, and T4, when the valve assembly associated with line **306** fully opens. On the other hand, valve assembly **260** (associated with line **308**) fully opens more quickly from an initial opening, as indicated by valve open time **318** representing an amount of time between T3, when the valve assembly **260** initially opens, and T4, when the valve assembly **260** fully opens. It will be understood that the time between T1 and T3 represents the initial phase of movement of valve assembly **260** (e.g., the closing the gap or energy storage phase). It will be understood that the time between T3 and T4 represents the subsequent phase of movement of valve assembly **260** (e.g., the energy release phase).

[0065] FIG. **9** is a sectional view showing another example of fluid applicator **200**. Fluid applicator **200**, in the example shown in FIG. **9**, includes a seal **238-2**, different than the seal **238-1** shown in FIG. **6**. Seal **238-2** includes a captivating wing **239** that contacts the inner surface **152** of retaining element **140** to create a friction fit between seal **238-2** and retaining element **140** to retain seal **238** and tip saddle **236** within spray tip assembly **230** when tip assembly is uncoupled from gun **110** (e.g., when coupling mechanism is unthreaded from the threads of gun **110**).

[0066] It will be understood that seal **238** (whether **238-1** or **238-2**) can comprise a non-metallic material, such as a hard polymer (e.g., hard plastic). In one example, seal **238** comprises nylon. In other fluid applicators, the seals, such as seal **138**, can comprise rubber. Seal **238** comprising a hard polymer (e.g., hard plastic), such as nylon, reduces the elasticity of the seal **238** and thus reduces the pressure accumulation of the volume downstream of the volume as compared to a seal, such as seal **138**, comprising a more elastic material, such as rubber.

[0067] FIG. **10** is a simplified pictorial illustration showing one example of a valve assembly **360** that can be used as alternative to valve assembly **260**. It will be understood that valve assembly **360**, like the valve assembly **260**, includes an additional spring (in addition to spring **170**), though the additional spring in valve assembly **360** shown in FIG. **10** is disposed in the valve fluid volume

112. The valve assembly **360** can be used in a fluid applicator, such as fluid applicator **100** (in place of valve assembly **160**) or such as fluid applicator **200** (in place of valve assembly **260**). It will be understood that, though not shown in FIG. **10**, valve assembly **360** further includes spring plate **168**, spring **170**, and seal **172**, and other items, as shown in FIG. **5**.

[0068] Valve assembly **360**, shown in FIG. **10**, includes valve seat **162**, valve ball **164**, valve shaft **366**, valve ball shaft **367**, additional spring **376**, and coupling mechanism **383**.

[0069] Coupling mechanism **383** couples valve shaft **366** to valve ball shaft **367** and thus, to valve ball **164**. Additional spring **376** is a tension spring that is tensioned to rest in, and return to, the state shown in FIG. **10**. Spring **376** is coupled between a shoulder **384** of coupling mechanism **383** and a shoulder **385** of valve ball shaft **367**. During an initial phase of movement (e.g., energy storage phase), when trigger **126** (not shown in FIG. **10**) is actuated in the direction **190**, valve shaft **366** (by virtue of connection to spring plate **168** as previously described) and thus, coupling mechanism **383** (by virtue of connection to valve shaft **366**) will move in the direction **190**. This movement will also cause spring **376** to stretch in the direction **190**, storing energy in spring **376**, until shoulder **386** of coupling mechanism **383** abuts (or contacts) shoulder **387** of valve ball stem **383**. During this initial phase of movement, the valve ball **164** is not dislodged from the valve seat **162**. Thus, it will be understood that the double spring system shown in FIG. **10** allows for an amount of expansion of the tension spring **376** before the valve ball **164** is dislodged from the valve seat **162**. That is, the double spring system shown in FIG. **10**, allows for an amount of movement of valve shaft **366** before dislodging of the valve ball **164**. The amount of movement is defined by the adjustable gap **378** between shoulder **386** and shoulder **387**. During a subsequent phase of movement (e.g., energy release phase), tension spring **376** will snap back to its tensioned state (releasing the stored energy), pulling and dislodging valve ball **174** from valve seat by driving movement of valve ball shaft **366** in the direction **190**, which assists in more quickly opening the valve.

[0070] FIGS. **11A-B** (collectively referred to herein as FIG. **11**) are sectional views showing one example fluid applicator **400**, illustratively a spray gun. Fluid applicator **400** can be used in a fluid application system, such as fluid application system **1** shown in FIG. **1**, in place of fluid applicator **10**.

[0071] Spray gun **400** includes gun (or gun body) **410**, inlet **420**, trigger **426**, spray tip assembly **430**, outlet **450**, and valve assembly **460**. Gun **410** includes, among other things, valve fluid volume **412**, inlet fluid volume **422**, handle **428**, valve contacting member **452**, and can include various other items.

[0072] Inlet **420** is coupled to a delivery line (e.g., **6**) and is configured to receive fluid pumped through the delivery line. Fluid enters gun **410** through inlet **420** and travels through inlet fluid volume **422**, disposed within handle **428**, to valve fluid volume **412**.

[0073] Spray tip assembly **430** includes spray tip **432**, outlet guard **434**, tip saddle **436**, and coupling mechanism **442**. Spray tip **432** can be similar to an used in place of spray tip **32**. Spray tip **432** includes tip piece **433** and pre-orifice piece **435**. In other examples, spray tip **432** could include, as a pre-orifice element, a pre-orifice formed in the body of spray tip **432** rather than a pre-orifice element as a separate pre-orifice piece, such as pre-orifice piece **435**. In other examples of spray tip **432**, a pre-orifice element, such as a pre-orifice piece **435**, need not be included. Tip piece **433** defines outlet **450** and is configured to emit fluid in a desired spray pattern. Spray tip **432** and tip saddle **436** are disposed within retaining element **440**. Spray tip **432** is rotatable within outlet guard **434** between a normal operating position (shown in FIG. **11**, where outlet **450** is facing away from gun **410**) and a cleaning operating position (where outlet **450** is facing toward gun **410**). Outlet guard **434** is coupled to coupling mechanism **442**, such as by a snap-on connection. Coupling mechanism **442** is also coupled to gun body **410** (threaded connection in the illustrated example) and thereby couples spray tip assembly **430** to gun **410**.

[0074] Additionally, it can be seen that tip saddle **436** is tapered and is disposed partially within the

outlet **481** of gun **410** and defines an inlet **482** of spray tip assembly **483**. The exterior surface of tip saddle **436** forms as seal between gun **410** and spray tip assembly **430**.

[0075] Valve assembly (or valve) **460** includes valve seat **462**, valve ball **464**, valve shaft **466**, and spring **470**, and can include various other items. Valve assembly **460** is moveable, in the directions **490** and **492**, by actuation of the trigger **426** (as described below), between a seated (or closed) position (as shown in FIG. **11**) and a plurality of open positions (from initially open to fully open). It will be appreciated by those skilled in the art that there is a range of open positions of valve assembly **460**, the range extending between an initial point when valve ball **464** is retracted sufficiently to initially allow some fluid through valve seat **462** to spray tip assembly **430**, though the valve ball **464** is still disrupting flow through the valve seat **462**, and a fully open (or fully retracted) point in which valve assembly **460** is retracted to the fullest extent and the valve ball **464** is fully pulled away from valve seat **462**. In other words, it will be understood that the valve assembly **460** can effectively control the flow rate of fluid to the fluid volume of spray tip assembly **430**.

[0076] Trigger **426** is moveably coupled to the gun body **410**, specifically to a valve contacting member **452**, and is moveable, such as by application and release (or reduction) of force from a user's hand, in directions **490** and **492**, between a deactivated (or resting) position (shown in FIG. **11**) and an activated position. Trigger **426** is coupled to valve contacting member **452** such that actuation of trigger **426** in the direction **490** causes rotation of valve contacting member **452**. As valve contacting member **452** rotates, from the actuation of trigger **426** in the direction **490**, valve contacting member **452** abuts and pushes against shoulder **468** of valve shaft **466** to move valve shaft **466** in the direction **490** and thereby compresses spring **470**, between valve shaft **466** and filter **469**, to open valve assembly **460**. When valve assembly **460** is opened, fluid flows from valve fluid volume **412**, through seat **462**, out of gun outlet **481** to spray tip assembly **430** through spray tip assembly inlet **483**.

[0077] Spring **470** biases valve assembly **460** in a closed position, and is compressible, as described above, to open valve assembly **460**. Hydrostatic pressure of fluid in valve chamber **412** also biases valve **460** in a closed position. The biasing force of the spring **470** and of the hydrostatic pressure can be overcome by application of force to trigger **426**.

[0078] FIGS. **12A-B** (collectively referred to herein as FIG. **12**) are sectional views showing one example fluid applicator **500**, illustratively a spray gun. FIG. **12A** is a side sectional view and FIG. **12B** is a top sectional view. Fluid applicator **500** can be used in a fluid application system, such as fluid application system **1** shown in FIG. **1**, in place of fluid applicator **10**. Fluid applicator **500** is similar to fluid applicator **400** shown in FIG. **11** and thus, similar items are numbered similarly.

[0079] It can be seen in FIG. **12** that fluid applicator **500** includes a valve assembly **560**. Valve assembly **560** differs from valve assembly **460** in that valve assembly **560** includes valve shaft **566**, different than valve shaft **466**, valve ball shaft **567**, stop **574**, and an additional spring **576**.

[0080] As illustrated in FIG. **12**, additional spring **576** is disposed around valve shaft **566** (and valve ball shaft **567**) and disposed between a shoulder **579** of valve shaft **566** and stop **574**. Stop **574** also serves to secure valve shaft **566** to valve ball shaft **567**, and thus valve ball **464** to valve shaft **566**. It can be seen that stop **574** is coupled to valve ball stem **567** (via a threaded connection in the illustrated example). As trigger **426** is actuated in the direction **490**, valve contacting member **452** rotates, as previously described. As valve contacting member **452** rotates, from the actuation of trigger **426** in the direction **490**, valve contacting member **452** abuts and pushes against shoulder **568** of valve shaft **566** to move valve shaft **566** in the direction **490**. During an initial phase of movement (e.g., energy storage phase) valve shaft **566**, moving in the direction **490**, slides along valve ball stem **467**, partially compresses spring **470** between valve shaft **466** and filter **469**, and compresses spring **476** between stop **574** and shoulder **579**, storing energy in spring **476**. During this initial phase of movement, the valve ball **464** is not dislodged from the valve seat **462**. Thus, it will be understood that the double spring system shown in FIG. **12** allows for an amount of

compression before the valve ball **464** is dislodged from the valve seat **462**. The amount of compression is defined by the adjustable gap (**578** shown in FIG. **13**) between shoulder **579** and stop **574**. During a subsequent phase of movement (e.g., energy release phase), as the trigger **426** continues to move in the direction **490**, the additional spring **576** will release the stored energy and drive movement of valve ball shaft **567** (and thus valve ball **464**) in the direction **490** (dislodging valve ball **464** from valve seat **462**) and will assist in opening the valve more quickly.

[0081] FIG. **13A** is a sectional top view of valve assembly **560**. FIG. **13B** is a perspective view of valve assembly **560**.

[0082] FIG. **14** is a schematic illustration of one example of valve assembly **560**. As shown in FIG. **14**, actuation of trigger **426** in direction **490** causes valve contacting member **452** to push against and drive valve shaft **566** in the direction **490**. During the initial phase of movement, valve shaft **566** compresses spring **470** and spring **576**, storing energy in spring **576**. In a subsequent phase of movement, spring **576** releases the stored energy, and drives movement of valve ball stem **567** (and thus valve ball **464**) in the direction **490** (dislodging valve ball **464** from valve seat **462**) and will assist in opening the valve more quickly.

[0083] FIG. **15** is a schematic illustration of one example fluid applicator **600** illustrating example operation of a valve assembly **660**. Fluid applicator **600** can be used in a fluid application system, such as fluid application system **1** shown in FIG. **1**, in place of fluid applicator **10**.

[0084] As illustrated fluid applicator **600** includes trigger **626** and valve contacting member **652**. Valve assembly **660** includes valve seat **662**, valve ball **664**, valve shaft **666**, spring **670**, and additional spring **676**. Trigger **626** is moveably coupled to a gun body (not shown in FIG. **15**) and is moveable, such as by application and release (or reduction) of force from user's hand, in directions **690** and **692**, between a deactivated (or resting) position (shown in FIG. **15**) and an activated position. As can be seen, in an initial phase of movement (e.g., energy storage phase), as trigger is actuated in direction **690**, spring **676** is compressed between shoulder **627** of trigger **626** and valve contacting member **652**, storing energy in spring **676**, until an adjustable gap **678** between trigger **626** (e.g., shoulder **629** of trigger **626**) and valve contacting member **652** is closed. During this initial phase of movement, the valve ball **664** is not dislodged from the valve seat **662**. Thus, it will be understood that the double spring system shown in FIG. **15** allows for an amount of compression before the valve ball **664** is dislodged from the valve seat **662**. The amount of compression is defined by the adjustable gap **678**. During a subsequent phase of movement (e.g., energy release phase), as the trigger **626** continues to move in the direction **690**, trigger **626** pushes against valve contacting member **652** which in turn pushes against shoulder **668** of valve shaft **666** and thus, drives valve shaft **666** and valve ball **664** in the direction **690** and compresses spring **670** against element **680**, opening the valve. During the subsequent phase of movement, the spring **676** releases the stored energy and assists in driving movement of the valve shaft **666** (and thus the valve ball **664**) in the direction **690** by assisting in driving valve contacting member **652** against the valve shaft **666**. Thus, the spring **676**, in the subsequent phase of movement, will assist in opening the valve more quickly.

[0085] It will be understood that element **680** could be a filter, similar to filter **469**, or a cap, similar to cap **173**, or could be another element, such as a gun body.

[0086] Though not shown in FIG. **15**, it will be understood that fluid applicator **600** can further include various other elements, such as elements of other fluid applicators shown herein, for example, but not by limitation, a spray tip (e.g. **32**, **132**, **432**, etc.) or spray tip assembly (e.g., **130**, **230**, **430**, etc.).

[0087] It will be understood that, in the examples of multi-spring valve assemblies **260**, **360**, **560**, and **660**, as the valve initially opens, pressure across the valve equalizes. The hydrostatic pressure, biasing the valve to a closed position, will be released, the release of which helps to allow the stored energy of the additional spring (e.g., **276**, **376**, **576**, or **676**) to open the valve more quickly. The resulting effect of the operation of the multi-spring valve assemblies **260**, **360**, **560**, and **660** is

a “snap open” of the valve. In the example of valve assemblies shown in FIGS. **16-19**, the release of a mechanical stop (e.g., detent mechanism, latch, etc.), rather than just a hydrostatic pressure, helps to allow the stored energy of the second spring to open the valve more quickly. In all of the example multi-spring valve assemblies shown herein, a biasing force (e.g., hydrostatic pressure, mechanical stop, etc.) is released that helps to allow the stored energy of one of the springs to be released to open the valve more quickly.

[0088] FIGS. **16-19** show example fluid applicators including a mechanical stop (e.g., detent mechanism, latch, etc.), that is releasable to help allow the stored energy of the second spring to open the valve more quickly. Advantageously, the use of a mechanical stop, such as shown in FIGS. **16-19**, allows for use of a stronger spring, as the second spring, than the second spring of previous examples as the mechanical stop can provide more resistance than hydrostatic pressure and thus allow for compression of a stronger spring. A stronger second spring will result in an even faster opening of the valve.

[0089] FIG. **16** is a schematic illustration of one example fluid applicator **700** illustrating example operation of a valve assembly **760**. Fluid applicator **700** can be used in a fluid application system, such as fluid application system **1** shown in FIG. **1**, in place of fluid applicator **10**.

[0090] As illustrated, fluid applicator **700** includes trigger **726** and valve contacting member **752**. Valve assembly **760** includes valve seat **762**, valve ball **764**, valve shaft **766**, spring **770**, and an additional spring **776**. Valve contacting member **752** includes a mechanical stop **782** (illustratively a spring-loaded detent mechanism) that includes a spring-loaded pin **784**. Trigger **726** is moveably coupled to a gun body **710** (only shown in part in FIG. **16**) and is moveable, such as by application and release (or reduction) of force from a user's hand, in directions **790** and **792**, between a deactivated (or resting) position (as shown in FIG. **16**) and an activated position. As can be seen, in an initial phase of movement (e.g., energy storage phase), as trigger **726** is actuated in the direction **790**, spring **776** is compressed between shoulder **727** of trigger **726** and valve contacting member **752**, storing energy in spring **776**, until an adjustable gap **778** between trigger (e.g., shoulder **729** of trigger **726**) and valve contacting member **752** is closed. During this initial phase of movement, the valve ball **764** is not dislodged from the valve seat **762**. Thus, it will be understood that the double spring system shown in FIG. **16** allows for an amount of compression before the valve ball **764** is dislodged from the valve seat **762**. The amount of compression is defined by the adjustable gap **778**. The mechanical stop **782** (by virtue of pin **784** being disposed in recess **711** of gun body **710**) provides resistance to the force of spring **776** such that valve shaft **766** is not moved. During a subsequent phase of movement (e.g., energy release phase), once the gap **778** is closed, with trigger **726** continuing to move in direction **790**, the additional force applied by trigger **726** of valve contacting member **752** will cause pin **784** to retract out of recess **711** and allow for movement of valve contacting member **752** and release of the stored energy of spring **776**. Thus, during the subsequent phase of movement, the spring **776** releases the stored energy and assists in driving movement of the valve shaft **766** (and thus the valve ball **764**) in the direction **790** by assisting in driving valve contacting member **752** against the valve shaft **766**. Thus, the spring **776**, in the subsequent phase of movement, will assist in opening the valve more quickly. During the subsequent phase of movement, spring **770** is also compressed against element **780**.

[0091] It will be understood that element **780** could be a filter, similar to filter **469**, or a cap, similar to cap **173**, or could be a part of gun body **710**.

[0092] Though not shown in FIG. **16**, it will be understood that fluid applicator **700** can further include various other elements, such as elements of other fluid applicators shown herein, for example, but not by limitation, a spray tip (e.g. **32**, **132**, **432**, etc.) or spray tip assembly (e.g., **130**, **230**, **430**, etc.).

[0093] FIG. **17** is a schematic illustration of one example fluid applicator **800** illustrating example operation of a valve assembly **860**. Fluid applicator **800** can be used in a fluid application system, such as fluid application system **1** shown in FIG. **1**, in place of fluid applicator **10**.

[0094] As illustrated, fluid applicator **800** includes trigger **826** and valve contacting member **852**. Valve assembly **860** includes valve seat **862**, valve ball **864**, valve shaft **866**, spring **870**, and an additional spring **876**. Valve contacting member **852** includes a mechanical stop **882** (illustratively a detent mechanism) that includes a pin **884**. Trigger **826** is moveably coupled to a gun body **810** (only shown in part in FIG. **17**) and is moveable, such as by application and release (or reduction) of force from a user's hand, in directions **890** and **892**, between a deactivated (or resting) position (as shown in FIG. **17**) and an activated position. As can be seen, in an initial phase of movement (e.g., energy storage phase), as trigger **826** is actuated in the direction **890**, spring **876** is compressed between shoulder **827** of trigger **826** and valve contacting member **852**, storing energy in spring **876**, until an adjustable gap **878** between trigger (e.g., shoulder **829** of trigger **826**) and valve contacting member **852** is closed. During this initial phase of movement, the valve ball **864** is not dislodged from the valve seat **862**. Thus, it will be understood that the double spring system shown in FIG. **17** allows for an amount of compression before the valve ball **864** is dislodged from the valve seat **862**. The amount of compression is defined by the adjustable gap **878**. The mechanical stop **882** (by virtue of pin **884** being disposed in recess **811** of gun body **810**) provides resistance to the force of spring **876** such that valve shaft **866** is not moved. During a subsequent phase of movement (e.g., energy release phase), once the gap **878** is closed, with trigger **826** continuing to move in direction **890**, pin **884** will retract out of recess **811** and extend into recess **837** of trigger **826**, which will allow for movement of valve contacting member **852** and release of the stored energy of spring **876**. Thus, during the subsequent phase of movement, the spring **876** releases the stored energy and assists in driving movement of the valve shaft **866** (and thus the valve ball **864**) in the direction **890** by assisting in driving valve contacting member **852** against the valve shaft **866** (e.g., against shoulder **868** of valve shaft **866**). Thus, the spring **876**, in the subsequent phase of movement, will assist in opening the valve more quickly. During the subsequent phase of movement, spring **870** is also compressed against element **880**.

[0095] It will be understood that element **880** could be a filter, similar to filter **469**, or a cap, similar to cap **173**, or could be a part of gun body **810**.

[0096] Though not shown in FIG. **17**, it will be understood that fluid applicator **800** can further include various other elements, such as elements of other fluid applicators shown herein, for example, but not by limitation, a spray tip (e.g. **32**, **132**, **432**, etc.) or spray tip assembly (e.g., **130**, **230**, **430**, etc.).

[0097] FIG. **18** is a schematic illustration of one example fluid applicator **900** illustrating example operation of a valve assembly **960**. Fluid applicator **900** can be used in a fluid application system, such as fluid application system **1** shown in FIG. **1**, in place of fluid applicator **10**.

[0098] As illustrated, fluid applicator **900** includes valve assembly **960**. Valve assembly **960** includes valve seat **962**, valve ball **964**, valve shaft **966**, spring **970**, additional spring **976**, spring plate **968**, spring clip **974**, mechanical stop **982** (illustratively a spring-loaded latch) that includes a latch plate **984**. Fluid applicator **900** can also include a trigger (not shown in FIG. **18**) moveably coupled to a gun body **910** (only shown in part in FIG. **18**) and moveable, such as by application and release (or reduction) of force from a user's hand, in directions **990** and **992**, between a deactivated (or resting) position (as shown in FIG. **18**) and an activated position. Further, the trigger is coupled to the spring plate **968** such that movement of trigger causes movement of spring plate **968**. As can be seen, in an initial phase of movement (e.g., energy storage phase), as the spring plate **968** is actuated in the direction **990** (by actuation of the trigger), spring **976** is compressed between shoulder **969** of spring plate **968** and spring clip **974** (shoulder **975** of spring clip **975**), storing energy in spring **976**, until an adjustable gap **978** between spring plate **968** (e.g., shoulder **969** of spring plate **968**) and spring clip **974** (e.g., an end of spring clip **974**) is closed. During this initial phase of movement, the valve ball **964** is not dislodged from the valve seat **962**. Thus, it will be understood that the double spring system shown in FIG. **18** allows for an amount of compression before the valve ball **964** is dislodged from the valve seat **962**. The amount of

compression is defined by the adjustable gap **978**. The mechanical stop **982** (by virtue of the spring-loaded resistance of latch plate **984** and the contact thereof with spring clip **974**) provides resistance to the force of spring **976** such that valve shaft **966** is not moved. During a subsequent phase of movement (e.g., energy release phase), once the gap **978** is closed, with the trigger continuing to move in direction **990**, an end **967** of spring plate **968** will contact latch plate **984** and cause latch plate **984** to move in the direction indicated by arrow **993**, which will allow for movement of spring clip **974** and release of the stored energy of spring **976**. Thus, during the subsequent phase of movement, the spring **976** releases the stored energy and assists in driving movement of the valve shaft **966** (and thus the valve ball **964**) in the direction **990** by assisting in driving spring clip **974** against the valve shaft **966** (e.g., against the shoulder **979** of valve shaft **966**). Thus, the spring **976**, in the subsequent phase of movement, will assist in opening the valve more quickly. During the subsequent phase of movement, spring **970** is also compressed against element **980**.

[0099] It will be understood that when the trigger is released and the valve assembly is correspondingly moved in the direction **992**, the mechanical stop will return to the position shown in FIG. **18**.

[0100] It will be understood that element **980** could be a filter, similar to filter **469**, or a cap, similar to cap **173**, or could be a part of gun body **910**.

[0101] Though not shown in FIG. **18**, it will be understood that fluid applicator **900** can further include various other elements, such as elements of other fluid applicators shown herein, for example, but not by limitation, a spray tip (e.g. **32**, **132**, **432**, etc.) or spray tip assembly (e.g., **130**, **230**, **430**, etc.).

[0102] FIG. **19** is a schematic illustration of one example fluid applicator **1000** illustrating example operation of a valve assembly **1060**. Fluid applicator **1000** can be used in a fluid application system, such as fluid application system **1** shown in FIG. **1**, in place of fluid applicator **10**.

[0103] As illustrated, fluid applicator **1000** includes valve assembly **1060**. Valve assembly **1060** includes valve seat **1062**, valve ball **1064**, valve shaft **1066**, spring **1070**, additional spring **1076**, spring plate **1068**, spring clip **1074**, mechanical stop **1082** (illustratively a spring-loaded latch) that includes a latch plate **1084**, mechanical stop **1096** (illustratively a spring-loaded latch) that includes a latch plate **1098**. Fluid applicator **1000** can also include a trigger (not shown in FIG. **19**) moveably coupled to a gun body **1010** (only shown in part in FIG. **19**) and moveable, such as by application and release (or reduction) of force from a user's hand, in directions **1090** and **1092**, between a deactivated (or resting) position (as shown in FIG. **19**) and an activated position. Further, the trigger is coupled to the spring plate **1068** such that movement of trigger causes movement of spring plate **1068**. As can be seen, in an initial phase of movement (e.g., energy storage phase), as the spring plate **1068** is actuated in the direction **1090** (by actuation of the trigger), spring **1076** is compressed between shoulder **1069** of spring plate **1068** and spring clip **1074** (shoulder **975** of spring clip **975**), storing energy in spring **1076**, until an adjustable gap **1078** between spring plate **1068** (e.g., shoulder **1069** of spring plate **1068**) and spring clip **1074** (e.g., an end of spring clip **1074**) is closed. During this initial phase of movement, the valve ball **1064** is not dislodged from the valve seat **1062**. Thus, it will be understood that the double spring system shown in FIG. **19** allows for an amount of compression before the valve ball **1064** is dislodged from the valve seat **1062**. The amount of compression is defined by the adjustable gap **1078**. The mechanical stop **1082** (by virtue of the spring-loaded resistance of latch plate **1084** and the contact thereof with spring clip **1074**) provides resistance to the force of spring **1076** such that valve shaft **1066** is not moved. During a subsequent phase of movement (e.g., energy release phase), once the gap **1078** is closed, with the trigger continuing to move in direction **1090**, an end **1067** of spring plate **1068** will contact latch plate **1084** and cause latch plate **1084** to move in the direction indicated by arrow **1093**, which will allow for movement of spring clip **1074** and release of the stored energy of spring **1076**. Thus, during the subsequent phase of movement, the spring **1076** releases the stored energy and assists in

driving movement of the valve shaft **1066** (and thus the valve ball **1064**) in the direction **1090** by assisting in driving spring clip **1074** against the valve shaft **966** (e.g., against the shoulder **1079** of valve shaft **1066**). Thus, the spring **1076**, in the subsequent phase of movement, will assist in opening the valve more quickly. During the subsequent phase of movement, spring **1070** is also compressed against element **980**.

[0104] Additionally, during the subsequent phase of movement, by virtue of movement of spring plate **1068** in the direction **1090**, latch plate **1098** will move in the direction **1099** and extend into recess **1065** of valve shaft **1066**. When trigger is released, an amount of movement of spring plate **1068** in the direction **1092** is allowed before valve shaft **1066** is moved in the direction **1092** as mechanical stop **1096** will resist the force of spring **1070** and thus, resist movement of valve shaft **1066** in the direction **1092**. Eventually, with continual movement of spring plate **1068** in the direction **1092**, spring plate **1068** will contact latch plate **1098** and latch plate **1098** will move in the direction **1093** and retract out of recess **1065**, allowing the stored energy of spring **1070** to release and close the valve more quickly, which will help to reduce spitting.

[0105] It will be understood that when the trigger is released and the valve assembly is correspondingly moved in the direction **1092**, the mechanical stop will return to the position shown in FIG. **19**.

[0106] It will be understood that element **1080** could be a filter, similar to filter **469**, or a cap, similar to cap **173**, or could be a part of gun body **1010**.

[0107] Though not shown in FIG. **19**, it will be understood that fluid applicator **1000** can further include various other elements, such as elements of other fluid applicators shown herein, for example, but not by limitation, a spray tip (e.g. **32**, **132**, **432**, etc.) or spray tip assembly (e.g., **130**, **230**, **430**, etc.).

[0108] FIG. **20** is a block diagram showing one example fluid application system **2000**. Fluid application system **2000** can include one or more pumps **2002**, a fluid source **2003**, a pump support structure **2004**, a fluid delivery line **2006**, a fluid applicator **2100**, and can include various other items **2012**, including, but not limited to, other items discussed or shown herein. Fluid applicator **2010**, itself, includes gun (or gun body) **2110**, trigger **2126**, spray tip assembly **2130**, valve assembly **2160**, and can include other items **2196**.

[0109] Gun (or gun body) **2110**, itself, can include an inlet **2120**, an outlet **2181**, and can include various other items **2197**. Inlet **2120** can be similar to any of the gun inlets described herein, such as inlet **120** or inlet **420**, or another type of gun inlet. Outlet **2181** can be similar to any of the gun outlets described herein, such as outlet **181** or outlet **481**, or another type of gun outlet. Other items **1097** can include any of the other items of guns (or gun bodies) described herein (e.g., **110**, **410**, **610**, **710**, **810**, **910**, **1010**, etc.) as well as other items.

[0110] Spray tip assembly **2130**, itself, can include spray tip **2132**, seal **2138** and can include various other items **2198**. Spray tip **2132** can be similar to any of the spray tips described herein, such as spray tip **32**, **132**, **432**, or another type of spray tip. As previously discussed, a spray tip defines an outlet of the applicator and can include a tip piece and a pre-orifice element. Further, as previously discussed, a spray tip can be reversible (e.g., rotatable about its longitudinal axis such that the outlet can face away from or towards the gun body). Seal **2138** can be similar to any of the seals described herein, such as seal **138**, seal **238**, the seal created by tip saddle **436**, or another type of seal. Various other items **2198** can include any of the other items of spray tip assemblies described herein (e.g., **130**, **230**, **430**, etc.) as well as other items.

[0111] Valve assembly **2160**, itself, can include one or more springs **2170**, and can include various other items **2199**. Springs **2170** can be similar to any of the springs described herein, such as spring **170**, or spring **170** and spring **276**, or spring **170** and spring **376**, or spring **470**, or spring **470** and spring **576**, or spring **670** and spring **676**, or spring **770** and spring **776**, or spring **870** and spring **876**, or spring **970** and spring **976**, or spring **1070** and spring **1076**, or other one or more springs. Various other items **2199** can include any of the other items of valve assemblies described herein

(e.g., **160, 260, 360, 460, 560, 660, 760, 860, 960, 1060**, etc.), as well as other items.

[0112] Trigger **2126** is moveably coupled to gun (or gun body) **2110** and is actuatable to open and close the valve assembly **2160**. Trigger can be similar to any of the triggers described herein (e.g., **126, 426, 626, 726, 826**, the trigger described in FIG. **18**, the trigger described in FIG. **19**, etc.).

[0113] Fluid applicator **2100** can include various other items **2196** including, but not limited to, any of the items of fluid applicators described herein (e.g., **10, 100, 200, 400, 500, 600, 700, 800, 900, 1000**, etc.). Pumps **2002**, in one example, can be similar to pump **2**, or can be other type of pumps. A fluid source **2003** can be a fluid container, such as a paint bucket (e.g., 5-gallon paint bucket, etc.) or another type of fluid container. Pump support structure **2004** can be similar to cart **4** or can be other types of pump support structures. Fluid delivery line **2006** can be similar to fluid delivery line **6** or can be another type of fluid delivery line. Pumps **2002** pressurize and pump fluid from fluid source **2003** and deliver the pressurized fluid to fluid applicator **2100** via fluid delivery line **2006**. The pumps **2002** can be supported by a pump support structure **2004**, such as a cart (e.g., **4**) or other pump support structure. Fluid application system **2000** can include various other items **2012**.

[0114] Thus, it can be seen that the examples herein provide for reducing spitting in fluid applicators by one or more of reducing size of the fluid volume downstream of the valve, reducing pressure accumulation of a volume downstream of the valve, or increasing the speed at which the valve can actuate (e.g., increasing the speed at which the valve can open or the speed at which the valve can close, or both). Each of reducing size of the fluid volume downstream of the valve, reducing pressure accumulation of a volume downstream of the valve, or increasing the speed at which the valve can actuate can reduce spitting.

[0115] Although the present invention has been described with reference to preferred examples, workers skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the invention.

[0116] Additionally, while a particular order of steps has been described for the sake of illustration, it is to be understood that some or all of these steps can be performed in any number of orders.

[0117] It should also be noted that the different examples described herein can be combined in different ways. That is, parts of one or more examples can be combined with parts of one or more other examples. All of this is contemplated herein.

Claims

1. A fluid application system comprising: one or more pumps; a fluid delivery line coupled to the one or more pumps; and a fluid applicator comprising: a gun; and a spray tip assembly removably coupled to the gun, the spray tip assembly comprising: a spray tip; a tip saddle; and a seal element disposed, at least partially, within the tip saddle and having a top surface in contact with an exterior surface of the tip saddle, the seal defining an inlet to the spray tip assembly and forming a seal between the gun and the spray tip assembly.
2. The fluid application system of claim 1, wherein the spray tip assembly further comprises: a retaining element, the tip saddle and the spray tip disposed, at least partially, within the retaining element; and wherein the seal element further comprises a captivating wing configured to contact an inner surface of the retaining element and thereby establish a friction fit between the seal element and the retaining element.
3. The fluid application system of claim 1, wherein the seal element comprises a hard polymer.
4. The fluid application system of claim 3, wherein the seal element comprises nylon.
5. A fluid application system comprising: one or more pumps; a fluid delivery line coupled to the one or more pumps; and a fluid applicator comprising: a spray tip assembly; and a gun, removably coupled to the spray tip assembly, the gun comprising: a trigger; and a valve assembly configured to be moved in a first direction by actuation of the trigger, the valve assembly comprising: a first

spring configured to bias the valve assembly in a second direction and to be compressed by movement of the valve assembly in the first direction; and a second spring configured to store energy in a first phase of the movement of the valve assembly in the first direction of movement and to release the stored energy in a second phase of the movement of the valve assembly in the first direction to assist in moving the valve assembly in the first direction.

6. The fluid application system of claim 5, wherein the second spring is a compression spring.

7. The fluid application system of claim 5, wherein the second spring is a tension spring.

8. The fluid application system of claim 5, wherein the gun further comprises: a spring chamber; a valve fluid volume; and a seal element configured to fluidically seal the spring chamber from the valve fluid volume.

9. The fluid application system of claim 8, wherein both the first spring and the second spring are disposed in in the spring chamber.

10. The fluid application system of claim 8, wherein the first spring is disposed in the spring chamber and wherein the second spring is disposed in the valve fluid volume.

11. The fluid application system of claim 5, wherein both the first spring and second spring are configured to contact fluid applied by the fluid application system.

12. The fluid application system of claim 5, wherein the second spring is in contact with the trigger.

13. A fluid applicator comprising: a gun comprising: a trigger; and a valve assembly configured to be moved in a first direction by actuation of the trigger, the valve assembly comprising: a first spring configured to bias the valve assembly in a second direction and to be compressed by movement of the valve assembly in the first direction; and a second spring configured to store energy in a first phase of the movement of the valve assembly in the first direction of movement and to release the stored energy in a second phase of the movement of the valve assembly in the first direction to assist in moving the valve assembly in the first direction; and a spray tip assembly removably coupled to the gun, the spray tip assembly comprising: a spray tip; a tip saddle; and a seal element disposed, at least partially, within the tip saddle and having a top surface in contact with an exterior surface of the tip saddle, the seal element defining an inlet to the spray tip assembly and forming a seal between the gun and the spray tip assembly.

14. The fluid applicator of claim 13, wherein the spray tip assembly further comprises: a retaining element, the tip saddle and the spray tip disposed, at least partially, within the retaining element; and wherein the seal element further comprises a captivating wing configured to contact an inner surface of the retaining element and thereby establish a friction fit between the seal element and the retaining element.

15. The fluid applicator of claim 13, wherein the second spring is one of a compression spring or a tension spring.

16. The fluid applicator of claim 13, wherein the gun further comprises: a spring chamber; a valve fluid volume; and a seal configured to fluidically seal the spring chamber from the valve fluid volume.

17. The fluid applicator of claim 16, wherein both the first spring and the second spring are disposed in in the spring chamber.

18. The fluid applicator of claim 16, wherein the first spring is disposed in the spring chamber and wherein the second spring is disposed in the valve fluid volume.

19. The fluid applicator of claim 13, wherein both the first spring and second spring are configured to contact fluid applied by the fluid applicator.

20. The fluid applicator of claim 13, wherein the second spring is in contact with the trigger.
